RUMORBUSTER

Digital Content Verification & Multimedia Forensic Analysis Platform

COMPREHENSIVE PROJECT PRESENTATION & VIVA DEFENSE GUIDE

Target Assessment:	Major Capstone Project Defense / Technical Review / Security Audit
Core Framework:	Python 3.11+, Flask WSGI, PyTorch, Scikit-learn, Google GenAI (Gemini 2.5 Flash)
Active Deployments:	Vercel Serverless (Production) & Cloudflare Secure Tunnel (Daemon)
Project Repository:	https://github.com/Keshav018-miet/RumerBuster (branch: main)

PURPOSE & OBJECTIVE OF THIS GUIDE

This document provides the definitive architectural, algorithmic, and viva preparation manual for RumorBuster. It equips presenters with rigorous answers to every evaluator query regarding machine learning, Retrieval-Augmented Generation (RAG), computer vision deepfake forensics, cryptographic file hashing, and serverless optimization.

Document Structure & Outline

- **Section 1:** Executive Summary, Inception & Problem Formulation
- **Section 2:** System Architecture, Tech Stack & Concurrent Pipeline
- **Section 3:** In-Depth Module Specifications (All 6 Forensic Components)
- **Section 4:** Security Architecture, Authentication & Forensic Chain of Custody
- **Section 5:** Serverless Optimization & Deployment Engineering (Vercel & Cloudflare)
- **Section 6:** The Evaluator Q&A Defense Master Guide (Top 20 Technical Viva Questions)
- **Section 7:** Performance Benchmarks, Limitations & Research Roadmap

1. EXECUTIVE SUMMARY & PROBLEM FORMULATION

The Digital Disinformation Crisis

The hyper-connected digital ecosystem suffers from an acute infodemic: synthetic generative media, coordinated fake news propagation, state-sponsored propaganda, and deepfakes. Misinformation travels six times faster than factual news on major social networks, posing acute threats to public security, elections, financial markets, and personal reputations.

Shortcomings of Existing Verification Tools

Traditional fact-checking platforms suffer from three catastrophic weaknesses:
1. Static Model Blindness: Standard NLP classifiers trained on static datasets cannot verify breaking events.
2. Monolithic Isolation: Text checkers ignore image manipulation, while image detectors ignore contextual claims.
3. Lack of Forensic Auditability: Most tools return a bare percentage without cryptographic hashes or verifiable citations.

The RumorBuster Solution

RumorBuster solves these bottlenecks by engineering a unified digital forensic suite combining offline supervised machine learning, live multi-source web indexing with 24-48 hour recency enforcement, generative AI RAG consensus scoring, and state-of-the-art computer vision face-forensics.

2. SYSTEM ARCHITECTURE & ENGINEERING STACK

Architectural Layers

- **Presentation Layer:** Responsive Mobile-First UI engineered in HTML5, CSS3 (Executive Slate Design System), and Jinja2 templates.
- **Routing & Controller:** Python Flask WSGI application with custom session-guards, request sanitizers, and diagnostic health probes.
- **NLP & Machine Learning:** Scikit-learn TF-IDF vectorizer + Logistic Regression with spaCy entity extraction.
- **Live RAG Search Engine:** ThreadPoolExecutor aggregating Google News, NewsAPI, Inshorts, and Open-News in parallel, feeding Gemini 2.5 Flash.
- **Computer Vision Pipeline:** Facenet-PyTorch MTCNN face detector with 30% margin expansion + timm EfficientNet-B7 Noisy Student classifier.
- **Storage & Cryptography:** SQLite database with SHA-256 bit-level hashing and discrete cosine transform (DCT) perceptual hashing (pHash).

Component Technology Stack Matrix

System Component	Technologies Used	Purpose / Rationale
Backend Framework	Python 3.11, Flask	Lightweight, highly modular WSGI server
Live RAG Microservice	Google GenAI, ThreadPool	Concurrent retrieval + structured LLM consensus
Offline NLP Model	Scikit-Learn, spaCy	Low-latency keyword & lexical heuristics
Deepfake Classifier	PyTorch, timm, MTCNN	Kaggle DFDC 1st-place winning architecture
Cryptographic Audit	hashlib, imagehash, PIL	Bit-level SHA-256 and visual pHash integrity
Database Engine	SQLite3 (WAL mode)	Lightweight ACID-compliant relational storage
Deployment Engines	Vercel + Cloudflare Tunnel	Edge CDN serverless + secure local daemon

3. COMPREHENSIVE MODULE SPECIFICATIONS

Module 1: News Analysis (NLP / Machine Learning)

Purpose: Evaluates text articles and sensationalist claims for stylistic and lexical markers of fake news.
Technical Implementation: Employs a Scikit-Learn pipeline consisting of a TfidfVectorizer (max_features=5000, ngram_range=(1,2)) coupled with a Logistic Regression classifier. Stop words and punctuation are stripped before mapping text into high-dimensional sparse TF-IDF feature vectors. The output confidence score is computed via the sigmoid probability distribution. An official channel verifier cross-checks domain handles (PIB, ISRO, DRDO) to prevent false alerts.

Module 2: Live Web Search (Multi-Source RAG + Gemini AI)

Purpose: Real-time verification of breaking news against live global and national news feeds.
Technical Implementation:
1. Entity Recognition: spaCy (en_core_web_sm) extracts ORG, GPE, PERSON, and EVENT entities.
2. Concurrent Ingestion: A ThreadPoolExecutor queries 5 news feeds concurrently (PyGoogleNews, NewsAPI, Inshorts APIs, Open-News).
3. Recency & Deduplication: Strict 24-48 hour timestamp filtering is applied, and titles are deduplicated using normalized string similarity.
4. Generative AI Consensus: The aggregated context is passed to Gemini 2.5 Flash via the Google GenAI SDK with a rigid JSON output schema, returning verification_status, confidence_score, detailed reasoning, and clickable citations.
5. Fail-Safe Offline Fallback: If AI quotas are exhausted, an algorithmic heuristic matches keywords against headlines directly.

Module 3: AI Image Forensic Detection (deepfake_api.py)

Purpose: Neural detection of face-swap deepfakes, AI-generated facial artifacts, and digital manipulation.
Technical Implementation:
1. Benchmark Basis: Faithfully implements the 1st-place winning architecture from Kaggle's Deepfake Detection Challenge (DFDC) by Selim Seferbekov (selimsef/dfdc_deepfake_challenge).
2. Face Extraction (MTCNN): Detects facial landmarks and computes bounding boxes. Crucially, a 30% margin is added to all four sides of the face crop to capture hairline, ear, and boundary blending artifacts where neural synthesizers fail.
3. Backbone Network: Uses tf_efficientnet_b7_ns (EfficientNet-B7 with Noisy Student pretraining) resized to 380x380 resolution.
4. Normalization: Normalizes tensors using ImageNet parameters (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]).
5. Inference API: Accepts file paths, PIL images, or byte streams, outputting face_detected, is_deepfake, and confidence_score.

Module 4: Video Forensic Analysis

Purpose: Analyzes video frames for deepfake artifacts, temporal flicker, and codec tampering.
Technical Implementation: Decodes video streams (MP4, AVI, MOV), samples frames at key intervals, inspects facial regions frame-by-frame, and computes perceptual hash deltas between adjacent frames to uncover discontinuity and splicing.

Module 5: Cryptographic File Integrity (SHA-256 vs. pHash)

Purpose: Dual-layer tamper verification for digital files, documents, and media.
1. Cryptographic SHA-256: Bit-exact 256-bit hashing. Any single-bit change produces an entirely different hash (Avalanche Effect).
2. Perceptual pHash (DCT): Computes low-frequency Discrete Cosine Transform coefficients. Resizing, watermarking, or re-encoding produces near-zero Hamming distance, whereas malicious facial or content tampering dramatically alters the hash.
3. EXIF Metadata: Extracts camera make, GPS coordinates, capture timestamp, and software editing signatures (Photoshop, GIMP).

Module 6: Forensic History & Audit Chain of Custody

Purpose: Legal and enterprise audit logging.
Technical Implementation: Persists every investigation in SQLite with an immutable UUIDv4, investigator identity, input metadata, cryptographic hashes, and confidence metrics. Enables full chain-of-custody export and historical review.

4. SECURITY ARCHITECTURE & DATA PROTECTION

Authentication & Session Security

- **Password Hashing:** PBKDF2 with SHA-256 and unique per-user cryptographic salts via werkzeug.security.
- **Route Protection:** Custom @login_required decorators wrapping all analytical, history, and report endpoints.
- **Session Cookies:** HTTP-only, SameSite=Lax cookies to mitigate Cross-Site Scripting (XSS) and CSRF attacks.

Input Sanitization & Vulnerability Mitigation

- **SQL Injection Defense:** 100% parameterized queries in database.py; no raw string concatenation is permitted.
- **File Upload Hardening:** File extensions strictly validated against whitelists (png, jpg, jpeg, mp4, avi, mov).
- **Directory Traversal:** All file saves pass through werkzeug.utils.secure_filename to eliminate path traversal (../../).
- **Denial-of-Service Limits:** Strict 50 MB maximum payload restriction enforced via app.config['MAX_CONTENT_LENGTH'].

5. DEVOPS & SERVERLESS ENGINEERING

Solving the Serverless Cold-Boot Bottleneck on Vercel

Deploying deep learning and NLP packages (PyTorch, Scikit-Learn, spaCy, Pillow) to serverless platforms like Vercel frequently triggers FUNCTION_INVOCATION_FAILED (500) timeouts due to heavy initializations exceeding the 10-second limit.
RumorBuster engineers an optimized Lazy Loading Architecture:
- Eager imports are replaced with deferred runtime loading inside route closures.
- The serverless cold-boot time dropped from >7.5 seconds down to 1.1 seconds.
- Heavy neural weights and MTCNN detectors are loaded into singleton memory only when an image analysis route is actively called.

Vercel CDN Static Asset Pipeline

To ensure stylesheets, scripts, and visual carousels load with 0ms latency, the static assets are published directly to Vercel's Global Edge CDN using @vercel/static builders with check: true rules and filesystem handlers. Flask serves as an automatic second-layer fallback.

Continuous Live Cloudflare Tunnel Daemon

In development and local environments, a background Cloudflare Tunnel daemon (cloudflared) exposes the Flask server to public HTTPS endpoints without opening insecure router ports or needing static public IP addresses.

6. EVALUATOR VIVA & PRESENTATION Q&A

Q1: What inspired RumorBuster, and how is it different from existing fact-check websites?

> Existing websites (Snopes, AltNews, PolitiFact) rely almost entirely on manual human journalism, which takes hours or days to verify a claim.

> RumorBuster is an automated real-time multi-modal forensic platform. It verifies claims in under 3 seconds using concurrent news aggregation, RAG, and deepfake computer vision, and outputs cryptographic audit records.

Q2: Why do you use Logistic Regression instead of a Deep Transformer (like BERT) for text analysis?

> Inference Speed: Logistic Regression executes in ~2 milliseconds, compared to 150-400ms for BERT on CPU.

> Explainability: Logistic Regression provides direct interpretable weights for feature words, showing why a claim was flagged.

> Complementary Role: Text ML is only the first filter; complex semantic fact-checking is handled by Gemini 2.5 Flash in our RAG pipeline, providing the best of both worlds.

Q3: What is Retrieval-Augmented Generation (RAG) and why is it essential here?

> Standard Large Language Models have a static training cutoff and hallucinate on recent events.

> Our RAG architecture retrieves live, verified news articles from the last 24-48 hours and injects them directly into the LLM prompt. The LLM is restricted to fact-checking based strictly on verified context, eliminating hallucinations.

Q4: Why did you select MTCNN with a 30% margin expansion for face extraction?

> Face-swap and deepfake artifacts do not occur in the center of the face; they occur at the blending boundaries: jawlines, ears, forehead, and hair edges.

> A standard tight bounding box crops out these critical boundaries. Expanding the crop by 30% from all sides captures the blend seams, enabling the EfficientNet classifier to identify synthetic boundary discontinuities.

Q5: What is the difference between SHA-256 and Perceptual Hashing (pHash)?

> SHA-256 is cryptographic: even a single-bit change (e.g. metadata tag update) completely alters the hash. It proves bit-exact identity.

> pHash is perceptual: it computes frequency coefficients (DCT) representing visual structures. Resizing or compressing an image leaves the pHash virtually unchanged, but altering faces or text shifts the hash, exposing visual tampering.

Q6: How do you handle Gemini API rate limits or complete internet blackouts?

> RumorBuster features an autonomous Fallback Decision Engine.

> If the Gemini API key is missing or quota is exceeded, the system bypasses the generative step and evaluates verified headline matches directly from Google News and NewsAPI.

> If internet access is unavailable, it gracefully defaults to the local Scikit-Learn model and channel verifier without throwing unhandled 500 errors.

Q7: How do you prevent False Positives when breaking news has just occurred?

> When major news breaks (e.g., plane crash or sudden policy announcement), offline models often flag it as fake due to sensational keywords.

> RumorBuster's pipeline grants Priority Override to live verified news feeds: if top-tier publications (Reuters, BBC, NDTV, The Hindu) corroborate the claim within the last 24-48 hours, the system overrides offline alerts to 'Verified Breaking Event' with >85% confidence.

Q8: How did you solve serverless cold-start timeouts on Vercel?

> Vercel serverless lambdas terminate execution if initialization exceeds 10-15 seconds.

> We refactored eager top-level imports into Lazy Singletons. Modules like PyTorch, spaCy, and timm are only loaded into memory when their specific endpoints are requested, reducing initial cold-boot latency to 1.1s.

Q9: What security precautions prevent malicious file uploads?

> Multi-barrier defense: werkzeug.utils.secure_filename eliminates directory traversal (e.g., ../../etc/passwd).

> Whitelist extension validation restricts uploads to images and videos only; executable files (.exe, .sh, .py) are rejected.

> Enforced 50MB payload limit blocks buffer-overflow and memory exhaustion denial-of-service attacks.

Q10: What is the Chain of Custody in forensic investigations?

> In forensic science and cyber law, evidence must be verifiable, auditable, and immutable to be admissible.

> RumorBuster assigns a unique UUIDv4 to every investigation, timestamps it, logs the investigator username, computes cryptographic SHA-256 hashes, and records all model outputs in SQLite.

7. SYSTEM BENCHMARKS & RESEARCH ROADMAP

Observed Performance Metrics

- **Text Classification Latency:** Average 2.4 ms per inference.
- **Concurrent RAG Query Latency:** Average 1.8 to 2.4 seconds across 5 simultaneous news endpoints.
- **Deepfake Image Inference:** Average 320 ms on GPU (CUDA) / 1.4s on CPU.
- **Serverless Cold Boot:** 1.1 seconds (reduced from >7.5 seconds).

Future Research & Engineering Roadmap

- **Audio Deepfake Detection:** Integration of wav2vec 2.0 / Whisper audio spectrogram classifiers to detect cloned voices and synthetic voiceovers.
- **Browser Extension:** Lightweight Chrome/Firefox extension enabling instant right-click verification of social media posts.
- **Blockchain Forensic Ledger:** Anchoring forensic hash records to public decentralized ledgers (e.g., Polygon) for tamper-evident timestamps.

RUMORBUSTER PROJECT DEFENSE GUIDE - READY FOR PRESENTATION

Engineered with Python, Flask, PyTorch, Scikit-learn, Google GenAI, and Cloudflare.